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CURRENT OPINION

Investigating the Effect of Return-to-Play Timing After Injury on Performance: Does the Analysis Answer the Research Objective?

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Abstract

Return-to-play decision making should be based on all the advantages and disadvantages of return to play for athletes, not just the risk of injury. For competitive athletes, this includes the effect of early versus delayed return to sport on performance. In this paper, we address the questions "How can I estimate the effect of injury on the individual's performance at return to play?" and "What is the effect of delaying return to sport on the individual's performance?". To address these questions, we describe (1) some foundational concepts, design and analytical challenges related to estimating the causal effect of return to play timing on performance in the athlete, (2) additional challenges if one is interested in the effects of delaying return to play and (3) differences when the questions relate to the team's performance. Although the analytical strategies described appear complicated, coaches and athletes make these judgements informally every day without explicitly stating their assumptions. Using a formal approach should help analysts provide the most valid answers to the questions asked by athletes and coaches. In brief, the choice of a comparison group depends on the research question and requires that one consider the hypothetical performance trajectory of the athlete had they never been injured. Thus, the optimal comparison group depends on the shape of the expected trajectory and the specific research question being asked.

Key Points

Choosing the comparison group for an observational study investigating the effect of injury on performance depends on the specific research question

Estimating the causal effect of injury on performance requires comparing what happened to performance after injury (which is observed and "factual") to what would have happened to performance if participants had not been injured (which is not observed and "counterfactual")

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1 Introduction

Return to play (RTP) after injury without adequate rehabilitation may lead to an increased injury risk compared with pre-injury status because of ongoing weakness [1] and decreased proprioception [2]. Although keeping an athlete out for longer periods of time may minimise the risk of reinjury, the athlete-patient is more than just a knee or shoulder injury. Return-to-play decision making should be based on an understanding of all the advantages and disadvantages of RTP for the athlete/team, not just the risk of injury. [3–5]

Injuries causing athletes to miss sports for more than a week remain relatively uncommon, with estimated rates of 1.8 injuries per 1000 athlete exposures for men's basketball, [6] 3.4 injuries per 1000 athlete exposures for women's hockey [7] and 5.9 injuries per 1000 athlete exposures for women's gymnastics [8] (calculated from non-time loss injury rates and the proportion of injuries with > 10 days of time loss). For competitive athletes, not allowing the athlete to participate in key competitions may affect the athlete's career, or their team's league standing. [9] Therefore, coaches and athletes must balance the consequences of early return (e.g. increased risk of injury, impaired performance because of ongoing injury), against the consequences of

delayed return (non-performance of an important player). [9] After all, athletes are playing a sport: "It's called football, not how many injuries can I prevent." (Caroline Bolling, personal communication). Using performance measures in addition to injury risk may also help motivate athletes and coaches to adopt primary injury prevention strategies. Although some studies have investigated the effects of injury on performance over one or more entire seasons subsequent to the injury (e.g. [10–13]), we are not aware of any study that evaluated the effects of injury on performance immediately at RTP.

In a previous paper, we outlined foundational concepts for studying the effects of injury rehabilitation on the risk of subsequent injury after returning to sport. [14] The purpose of this paper is to extend and adapt these methods to solve additional challenges when estimating the effect of RTP timing on performance in competitions. This should help coaches and athletes make more informed RTP decisions. We address the following questions: "How will the injury affect the individual's performance when they RTP?" and "What is the effect of delaying RTP on the individual's performance?". To answer these questions, we describe (1) some foundational causal concepts, design and analytical challenges related to estimating the causal effect of injury on performance immediately at RTP, (2) additional challenges if one is interested in how delaying RTP would affect performance and (3) differences when the questions relate to the team's performance.

We use alpine skiing as our example for individual sports, and American Football as our example for team sports. For simplicity in this conceptual paper, we consider predicted performance as a composite measure that uses a continuous scale (e.g. 0–100), such as the Player Efficiency Rating in the National Basketball Association (https://www.basketball-reference.com/about/per.html) or international tennis Elo ratings (http://on-the-t.com/ratings/atp-singles/). Unless otherwise stated, performance refers to results in competitions rather than training. The same methods can be used for binary outcomes (win-lose) and the outcome simply represents the probability of success under different RTP decisions.

2 Foundational Causal Concepts

A key methodological assumption for determining causal effects of injury on any outcome is that the groups being compared should have similar expectations of averages or probabilities for the outcome if both groups received the same treatment. [15] Similar to our previous reports on recurrent injuries, [16] we must carefully design our studies if we want to interpret the results as causal effects of injury on performance immediately at RTP. For simplicity, we assume that everyone who is active at the beginning of the season is uninjured. In brief, consider that in every sport, different participants have different performance abilities before the start of the season owing to their talent, fitness and role. The participants who become injured earlier are quite possibly different in talent, fitness and role compared with the participants who become injured later or not at all. The optimal way to estimate the total effect of injury on performance immediately at RTP is to compare their performance after injury (called "factual" in causal inference because we observe it) to how they would have performed had they never been injured (called "counterfactual" because we cannot observe it). As we cannot collect data on how they would have performed if never injured, we must infer the "counterfactual" data from data collected on an appropriate comparison group, as outlined in the subsequent sections. Similar to our previous papers, we believe the most useful approach is often a multi-state framework in which participants move from one state to another, and we measure the difference in performance across particular states (which may or may not involve multi-state statistical models). [17]

3 Effect of Injury on Individual Performance: Choosing the Optimal Comparison Group

When assessing individual performance, the correct comparison group depends on how performance changes over time. Figure 1 illustrates three theoretical trajectories of performance over the course of a season for an athlete who never gets injured (a very simplified example for pedagogical/educational purposes). The solid line indicates the performance trajectory for an athlete whose performance remains constant throughout the season (constant performance athlete). The dashed line indicates the performance trajectory of an athlete whose performance increases throughout the season (increasing performance athlete). This might occur in a young athlete who is learning rapidly through training, or because they recently began competing against higher level athletes. The dotted line indicates the performance trajectory for a decreasing performance athlete. This might occur in an older athlete who already understands the sport very well, and cumulative fatigue occurs over the season because of an inadequate recovery time. Although we do not know what the true trajectory is for an individual athlete, determining the effect of injury on performance requires that we estimate the trajectory using an appropriate comparison group. We discuss each of these contexts separately.

Consider the context of the constant performance alpine skier where the season begins in September, the athlete becomes injured in November and returns to sport competition in early January (Fig. 2A). Had they not been injured (solid black line), their performance would have remained

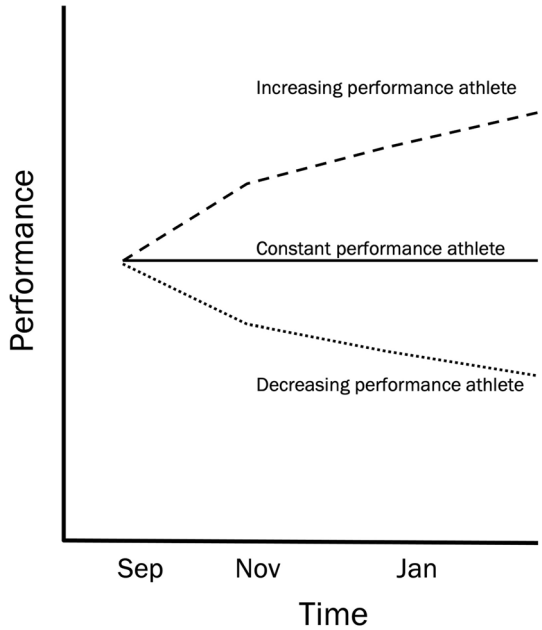

Fig. 1 Three hypothetical trajectories of performance measured on a continuous scale over time during a single season for athletes who remain uninjured. Athletes may increase their performance over time (dashed line), have stable performance over time (solid line) or have decreasing performance over time (dotted line)

constant. The grey line represents an "injury-recovery" curve. Immediately after injury, their potential performance if they were to RTP and compete decreases substantially. As time passes and recovery occurs (with or without formal rehabilitation), the athlete has less pain, more strength, power and endurance, and their potential performance increases. The exact shape of the injury-recovery curve will affect our estimates for the effect of injury on performance, but it does not affect the analytical strategies, and the concepts described below apply to any shape of the curve. Table 1 summarises the data one might require to study the effects of injury on performance immediately at RTP.

In Fig. 2A, the alpine skier now returns to sport before the injury-recovery curve has reached their baseline level of performance. The causal effect of injury on performance immediately at RTP is simply the difference in performance between what would have happened if they were never injured (unobserved) and their actual performance upon RTP when they were injured (observed). We can calculate this difference (1) for each day after return to activity (simple difference in performance) or (2) sum the effect across all days (light grey area between the two "performance" trajectories).

Our challenge is to pick the optimal comparison group to estimate what would have happened to performance if the alpine skier was never injured. If we assume the skier is a constant performance athlete, we could compare their performance level after return to sport with the same athlete's performance at baseline, or just prior to injury because it is mathematically equivalent to baseline. Alternatively, we could compare the athlete's performance after injury with athletes who had similar levels of performance prior to injury, if these athletes were also expected to have constant levels of performance throughout the season.

This analysis would generally only consider the effect of injury on performance after returning to sport and ignores that the alpine skier missed competitions. For some questions, we might want to estimate the total effect of injury on performance. Because the alpine skier cannot obtain any points toward World Cup standings while injured, we might consider assigning a performance = 0 for competitions that were missed. Alternatively, some sports might rate athletes based on an average of their results in competitions they participated in. In these cases, not competing leaves the rating untouched and a poor performance because of injury would lower the ratings. In these contexts, we would use some type of weighting scheme that accounts for the advantages and disadvantages of competing.

Figure 2B illustrates an improving performance athlete. In this context, comparing performance to baseline in September (light grey area) would grossly underestimate the effect of injury on performance because performance would have increased throughout the year had the skier not been injured. Some might suggest that we should compare the performance after injury to the performance just before injury in November (sum of light and dark grey areas). However, we would again be underestimating the effect of injury on performance because performance in January would have been better than the performance in November had the athlete never been injured. Rather, Fig. 2C illustrates the effect of injury on performance when we use a comparison group that represents the same performance that the skier would have had if they had never been injured (sum of light grey, dark grey and black areas). Because we never observe the performance of our skier if they were uninjured, we must estimate the performance only from other skiers. Our analytical strategy should therefore match the injured skier to other skiers who have performance trajectories similar to the injured skier's theoretical uninjured trajectory. Similar to the constant performance athlete, we might use a weighting scheme to account for the advantages and disadvantages of competing when (1) not competing has no effect on standings and (2) a poor performance has a detrimental effect on competitive standing.

Figure 3 illustrates that the same principles apply for the skier with decreasing performance. However, if the rest that occurs following the injury leads to an injury-recovery curve where the skier regains the level of performance they had at baseline, the injury (or more precisely, the rest that occurs because of injury) might improve performance on return to

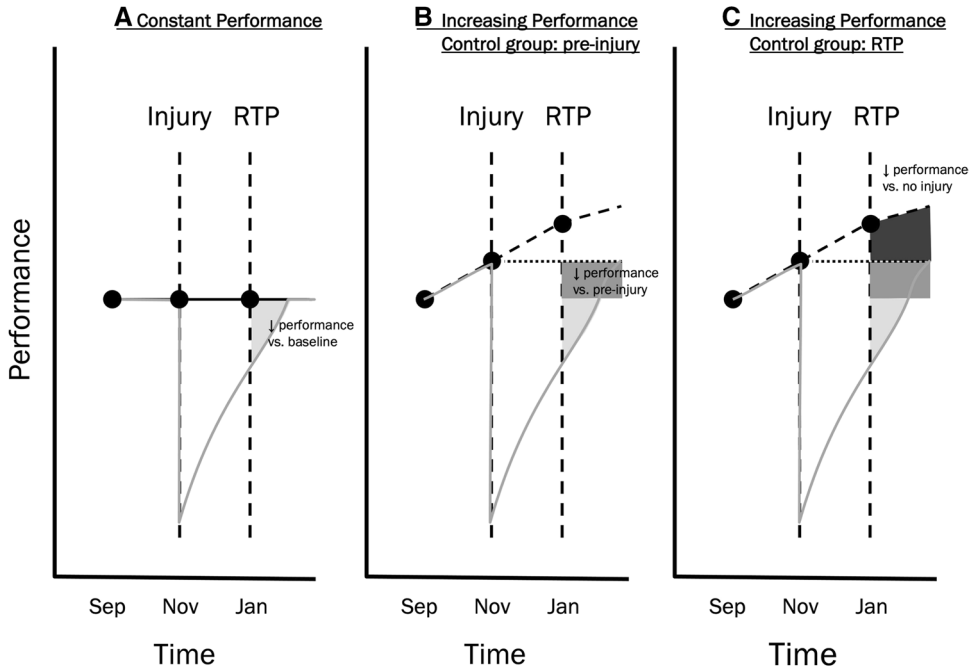

Fig. 2 Hypothetical trajectories of an athlete who becomes injured in November and returns to sport in January. In **A**, the athlete's theoretical performance would have remained constant had they not been injured (solid black line). The theoretical performance after injury (grey line, "injury-recovery" curve) decreases with injury and then gradually increases during recovery. The effect of injury on performance (grey shaded area) is the difference between the theoretical performance had they not been injured (not observed) and their performance on return to play (RTP). In **B**, the athlete's theoretical performance had they never been injured increases throughout the season. In this context, the control group should represent what the performance of the athlete would have been at the time of RTP had they never been injured. Panel **B** illustrates the detrimental effect on performance if we compare to pre-injury levels. This includes (1) the area from **A** and (2) a detriment on performance compared with their immediate pre-injury state (dark grey area). **C** Illustrates that there is an additional detriment on performance when we compare their observed performance to what their performance would have been had they never been injured (black area). The performance had they never been injured is estimated from a control group that followed the same trajectory prior to injury

sport (white cross-hatched area). If we consider the total effect of injury over the entire date range, there is a large decrease in performance between November and January when the skier is not participating.

4 Additional Challenges when Estimating the Effect of Delayed RTP on Performance

The approach described in the previous section estimated the effect of injury on performance immediately after RTP. However, a different and important practical question for every injured athlete is whether the athlete should be returned to sport today, a week from today or at some other time. To address this question, we still compare our observed performance to a theoretical unobserved performance, but the unobserved performance is now more difficult to estimate.

Figure 4A illustrates an increasing performance athlete where injury negatively affects performance. The same principles apply for constant and decreasing performance athletes. Because we are interested in estimating the (counterfactual) performance if we returned the skier later in the season compared to when they did return, our comparison group for the counterfactual data must represent what would have happened to the skier had they been injured and returned to sport later. Therefore, we can no longer use uninjured skiers as the comparison athletes. In Fig. 4A, returning the skier to sport alters the injury-recovery curve (from grey line to dotted line) because there is more time for recovery of strength, proprioception and technique. We assume the increased physical capacities greatly outweigh any decreased technical skill or timing for this simplified example. Therefore, theoretical performance after delayed RTP is better each day during this recovery compared with their observed performance when returning to sport earlier (white cross-hatched area). However, the skier and coach also realise that the skier is out of competition longer, and they want to include this negative effect for a lack of performance (grey area). The injury-recovery curve in Fig. 4A is just an example, and the analytical principles described are valid regardless of its shape. For some research questions,

Table 1 General recommendations for data gathering and analysis when estimating the effect of RTP timing after injury on performance

Study design	Main summary point	Notes
Case–control studies	Any athlete that is a case must still be eligible as a control prior to their injury	Many studies classify participants as cases if they ever developed an injury, and choose controls from those that were never injured. This increases the risk of bias. If the data were collected prospectively, the performance trajectory data in the months September to November for an athlete injured in December would still be used as a control to understand performance trajectory of uninjured players between September and November. Not allowing for these time-dependent effects may introduce bias in case–control studies
Constant performance athlete	An appropriate comparison group may be the same athletes prior to injury, with caveats	As the performance does not change over time in the uninjured state, we can use the athletes' own baseline data as the comparison group. However, there are challenges when injuries occur very early in the season because there may be very limited "comparison" data available. In these cases, one might use performance data from the same athletes the year before, if it appears reasonable that these represent a good measure of their performance early in the subsequent season. Alternatively, one might use performance data from other athletes with similar performance trajectories
Increasing or decreasing performance athlete	An appropriate comparison group requires athletes with similar performance trajectories prior to the injury, with caveats	This comparison group requires detailed data on different types of performance measures in all athletes. This allows us to match the comparison group based *on performance* prior to injury and after injury, so we can predict what the performance would have been after injury if no injury had occurred. This contrasts with a commonly used approach where investigators match on non-performance factors believed to *predict performance* such as draft position, player position, body mass index and seasons played. However, one would expect the actual historical performance data to much better predict future performance compared with models that are limited to non-performance metrics. For simplicity, we assume in the main text that one can match the comparison group based on performance just prior to injury. Although this is one strategy, a better strategy might be to match on the trajectory of performance from the beginning of the season until the time of injury. The choice would depend on the sample size (will have to limit matching criteria if small sample size) and data available
Estimating effect of delayed RTP	An appropriate comparison group requires matching for performance characteristics during rehabilitation phase (i.e. after injury and before RTP)	The comparison group needs to reflect what the athlete's performance would have been had they RTP at two different times after injury. Therefore, all of the participants in the comparison group must be injured. We then choose athletes with similar injuries and prognosis, which requires detailed data on their injury at the time of injury. In addition, some athletes might progress rapidly through rehabilitation and others have a slower course or exacerbations. The better we can match the trajectories throughout rehabilitation of our injured group to the comparison group, the more accurate our estimate will be. This level of detail at a research quality level is difficult to obtain, and some simplifying assumptions will likely have to be made and justified in any particular study

Table 1 (continued)

Study design	Main summary point	Notes
Team sports	Requires considerations of similarity between replacement players and injured players prior to the injury even though the replacement player may never have played during that time. One should also consider adjustments for team standing and style of play	The trajectory of performance in replacement players is generally not available because they are either players brought up from lower leagues (which would require having data on performance at lower league levels) or players who are now required to play a different position or style (would require data that indicate past performance at these new positions or styles). Investigators should try to limit their assumptions as much as possible and explain why the assumptions made are reasonable given the context of the research question

RTP return to play

the overall effect on the skier might be a simple sum of the grey area and white cross-hatched area. In contexts where poor performance in competitions causes an athlete's ranking to decrease more than not competing at all, one would use some form of weighting as mentioned in the previous section.

To estimate this theoretical recovery curve if we delayed return to sport, we follow similar principles to any "prognostic" research question. We know the skier's baseline measures and their injury state at the time of RTP (pathology, functional measures). We then select similar skiers who were injured and estimate how their performance changed over time when they RTP. If some of these skiers returned to sport later than our skier for the same original pathology, we can use their performances to help estimate the theoretical curve for our injured skier if RTP were delayed. The same approach can be used to estimate the effect of returning a skier to sport earlier. In this case, we would generally use our skier's injury state from an earlier timepoint (e.g. 1 week earlier) and match it to other injured skiers with similar characteristics and injuries.

5 Additional Considerations for Team Sports

Figure 4A describes an approach for individual athletes. In team sports such as American Football, owners, coaches and fans may be more interested in the performance of the team. An injured athlete may be replaced by another athlete. The coach needs to decide if they should allow the injured athlete to play, or use an uninjured athlete as a replacement. The same principles can be applied, but we must now consider the performance of the team, rather than the individual athlete.

Although one could simply replace the above graphs with "team performance" measures, examining the theoretical performance level of the injured and replacement athletes yields important insights. Figure 4B illustrates an example where the replacement athlete's uninjured performance trajectory throughout the year (solid black line) increases over the season and has the same general shape as the injured athlete had they not been injured, but at a much lower level. In this example, the healthy replacement athlete in January does not even perform as well as the recovering injured athlete in January. Therefore, if we delay return to sport for the injured athlete, the replacement athlete performance is not as good as that of the injured athlete, and the team performance would be expected to be lower. However, when the injured athlete does return (cross-hatched area), the injured athlete is much better than they would have been had they returned earlier. Therefore, the team's performance later in the season would be better with delayed return. It is the balance of these factors that makes coaching decisions so difficult, as

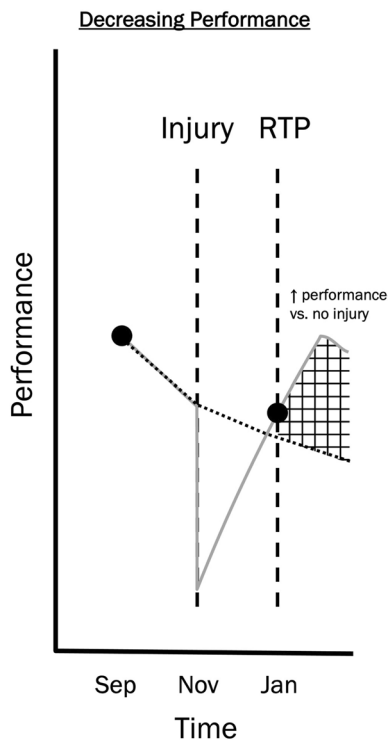

Fig. 3 An athlete's theoretical performance deteriorates over the course of the season. In this context, the athlete's opportunity to rest may allow them to recover some of their earlier capacities so that their overall performance on returning to play is better than it would have been had they never been injured. *RTP* return to play

illustrated by the recent controversy of whether Baker Mayfield (quarterback) of the National Football League Cleveland Browns should have continued to play injured or taken some time off to recover (https://clutchpoints.com/browns-news-the-miraculous-truth-on-baker-mayfields-injury/).

Figure 4C illustrates the context when the replacement athlete is better than the injured athlete playing injured, but not if the injured athlete were healthy. The decision is much easier in the Fig. 4C example because there is only benefit to the team's performance regardless of whether the injured athlete has an early or delayed return to sport.

The decision must also be made in the context of other factors such as team standing and playoff contention. For example, should James Harden have returned to play in Game 5 of the Eastern Conference semi-final series in the 2021 National Basketball League's playoffs? (https://www.espn.com/nba/story/_/id/31640743/brooklyn-nets-james-harden-hamstring-back-action-game-5-vs-milwaukee-bucks). If a team is fighting for a playoff spot or in the playoffs, then returning the injured athlete sooner might be appropriate as a poor team performance could mean missing the playoffs altogether.

6 Discussion

Analysing the effect of injury on performance requires comparing observed performance after injury to what performance would have been had the athlete never been injured (unobserved). We can estimate the unobserved performance trajectory by selecting a group of athletes who are similar to the injured athlete, but who were never injured. Estimating the effect of (1) delaying return to sport on individual performance or (2) using a replacement athlete in a team sport during the delayed return to sport follows the same principles but choosing the appropriate comparison group is more difficult.

Real-world data originate from contexts that are much more complex than our simple vignettes. Our examples are meant only as a framework for how to design a study or analyse data. The trajectory of performance may be different for different measures of performance (e.g. gymnastics score on an apparatus, probability of winning a medal in skiing). This highlights the importance of understanding the sport and context in choosing the most appropriate measure. Knee medial collateral ligament injuries are likely to have different healing trajectories than shoulder ligament injuries on average. Two athletes with similar medial collateral ligament injuries and the same tissue strength state may have different performance trajectories because (1) proprioception is affected in one injury but not the other or (2) the two athletes play different sports with different demands on the knee. Therefore, the decision of which injury states to use for matching within a state-based approach [17] is also more complicated than described in our simple vignettes. Although these contexts are more complex, the principles we have outlined remain valid.

In general, performance metrics are either non-standardised statistics (e.g. goals, yards after a catch, race time) or model-standardised metrics. Model-standardised metrics include three broad categories (1) model-weighted counting statistics such as Wins above Replacement, which reweights observed counting statistics and standardises the production value of a player against the production value of a replacement, (2) model-adjusted statistics, where a player's contribution towards counting statistics are adjusted for contextual factors by an explicit model, like teammates in the regression-adjusted plus/minus framework [18] and (3) residualising approaches, where the outcome itself is derived from the output of a model. For example, one might measure player contributions through actions or spatial roles towards a model outcome such as the expected value of a possession in team sports. [19] Player contributions towards these model outcomes can then be further adjusted just like counting statistics to separate contextual factors from a player's individual performance. In general, good performance

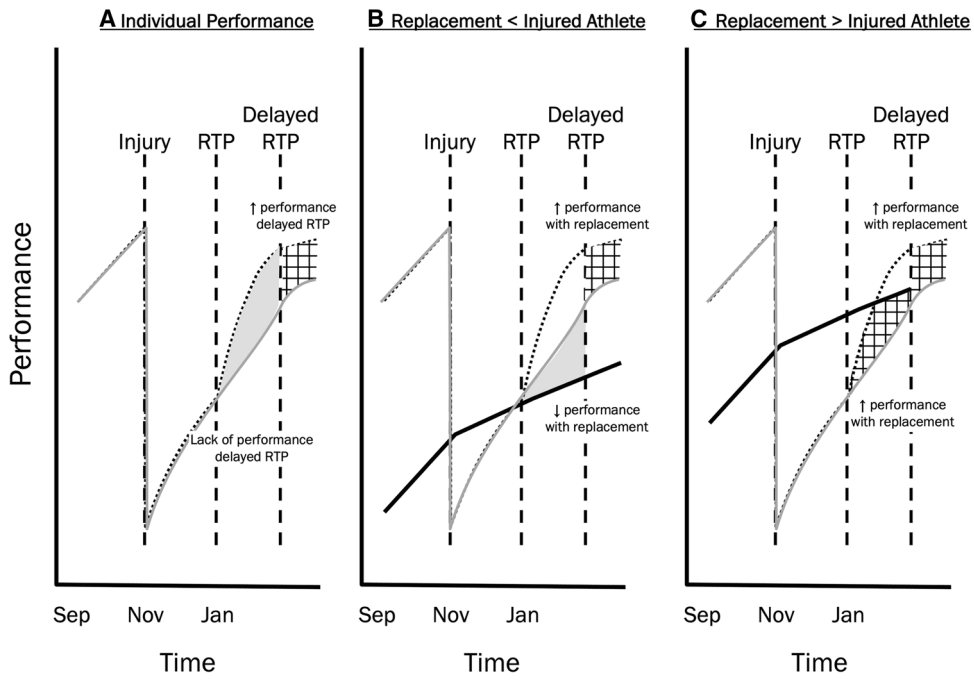

Fig. 4 Hypothetical trajectories for performance with a standard time to return to play (RTP) or a delayed return to play (delayed RTP). In **A**, the athlete's observed performance (grey injury-recovery curve after RTP) is compared to what the theoretical performance would have been had they only returned to play later (dotted line beginning at RTP). The effect on performance after the delayed RTP is beneficial (white cross-hatched area) because of the additional recovery time. If one wanted to consider the time the athlete was not playing between RTP and delayed RTP, the grey area between the dotted line and grey line would be considered as detrimental effects on performance. In **B**, the black solid line represents the performance trajectory of a replacement athlete who is not expected to perform as well as the injured athlete playing injured. If the team replaces the injured athlete with this low performance athlete, the team's performance will be reduced during the time between RTP and delayed RTP. However, the injured athlete's performance (and presumably the team's performance) would be much improved once they return (white cross-hatched area). In **C**, the replacement athlete (black solid line) has a higher predicted performance than the injured athlete playing injured, but a lower predicted performance than the injured athlete in a healthy state. In this case, the team performance is increased between RTP and delayed RTP, and again after delayed RTP

measures for estimating the effect of injury will isolate the performance and ability of individual players and context, but this may be more challenging in some sports than others depending on the nature of the game and the availability and abundance of high-information outcomes.

In this paper, we used a continuous measure of performance (e.g. race time). How to measure performance is a complex construct. Raysmith et al. have described some of the foundational principles for athletics, [20] and in a more general framework. [21] The same methods can be used for binary outcomes (win-lose) but the y-axis would represent the difference in the probability of success under the different performance trajectories.

The main questions in our paper refer to the effects of injury on performance immediately at RTP, and how to assess the risks and benefits of returning a player to sport sooner rather than later. We are not aware of any study that has attempted to answer these questions, partly because it requires a much greater level of detailed game-by-game data (both performance and injury related) than is usually available (Table 1). That said, there are studies evaluating the long-term effects of certain injuries in specific sports beyond just counting the number of games played after injury. Although a full review of all such studies is beyond the scope of this article, we comment on a few examples below.

In a few case–control analyses in professional basketball [10–12] and football [13], the authors studied the effect of injury on performance in one or more *full seasons after RTP*. All of these studies used counting statistics (e.g. NBA Player Efficiency Rating, football defensive players ratings, games started and games played). In these studies, whether RTP was one or several weeks earlier or later was not relevant to their research question. Case–control analyses are more complicated than they might appear. For example, in some of the articles, authors matched cases to controls on several pre-injury variables but did not include these matching variables in the statistical model. This increases the risk of bias. [22, 23] Further, in the papers using an external comparison group, bias is likely because cases were not eligible to be part of the control group prior to their injury. [24] A player who injures themselves in the fifth year of their career would

be a case for the fifth year, but should also be eligible as a control for players who are injured in their first few years of their career. Otherwise, by definition, we are increasing the difference in risk of injury between the cases and controls, which would increase the bias if the risk of injury is related to potential performance later in the season. There is also an historical cohort study examining performances pre-post hip arthroscopic surgery across a wide range of elite sports. [25] However, the authors did not use a control group to estimate declining performance over time because of reasons other than injury.

7 Conclusions and Perspective

We have illustrated how one can approach answering two important questions coaches and athletes ask clinicians: "How will the injury affect the individual's performance when they RTP?" and "What is the effect of delaying return to sport on the individual's performance?" We also illustrated additional challenges when one is interested in delaying RTP or in team sports. Although the analytical strategies described appear complicated, coaches and athletes make these judgements informally every day without explicitly stating their assumptions. For example, some may compare the athlete's performance to their pre-injury state rather than the state that would have occurred had they never been injured. Using a formal approach should help analysts provide the most valid answers to the questions asked by athletes and coaches.

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Declarations

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